CULTURAL PRESERVATION · NON-COMMERCIAL

NaraMotion

Low-cost motion capture and AI coaching for Indonesian traditional dance

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naramotion.web.app



01 · THE PROBLEM

Dance is passed on body to body, and video keeps only the surface

Knowledge is lost

Joint angles, timing and finger shapes taught by masters disappear when only video is archived.

Capture is costly

Optical studios such as VICON are out of reach for schools and sanggar, so almost no Indonesian dance has motion data.

Feedback is scarce

Students practising alone cannot tell which joint is wrong; a teacher is needed for every correction.

02 · THE SOLUTION

One affordable system to capture, recognise and teach

Capture

A 10-node GY-85 9-DoF suit on ESP32-C3 boards, or just a phone camera.

Recognise

AI trained on 2,568 Balinese videos names 24 basic movements in the browser.

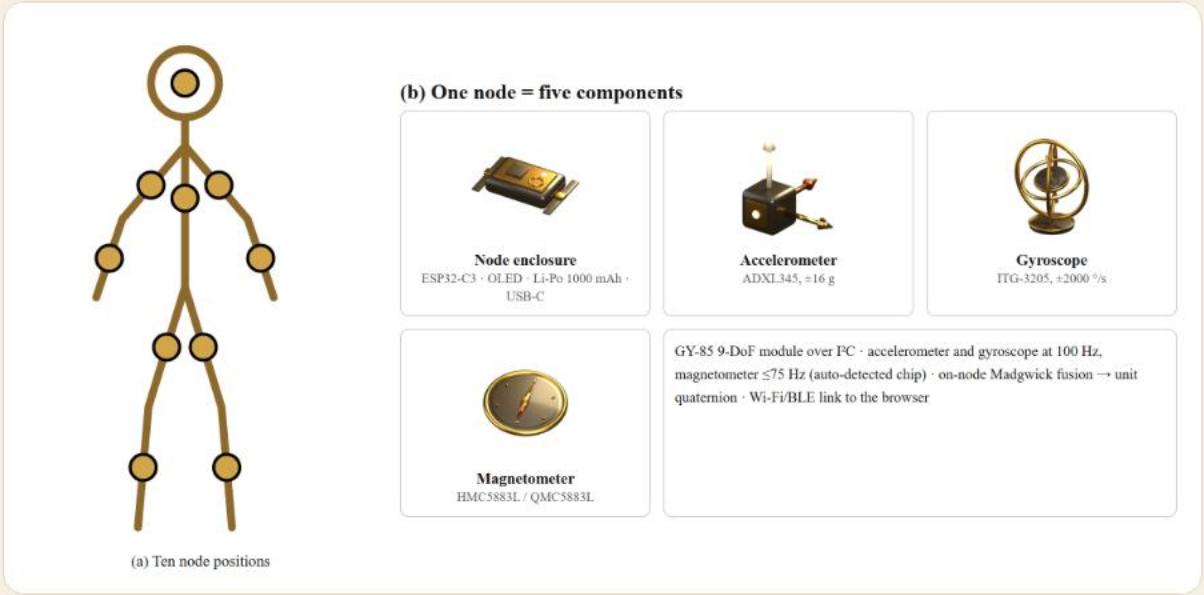
Teach & archive

Studio, Kelas Tari and Arsip give scores, joint feedback and 3D references.

Static web app · Firebase · free Kaggle GPUs for training

03 · HARDWARE

Ten 9-DoF nodes cover every limb segment

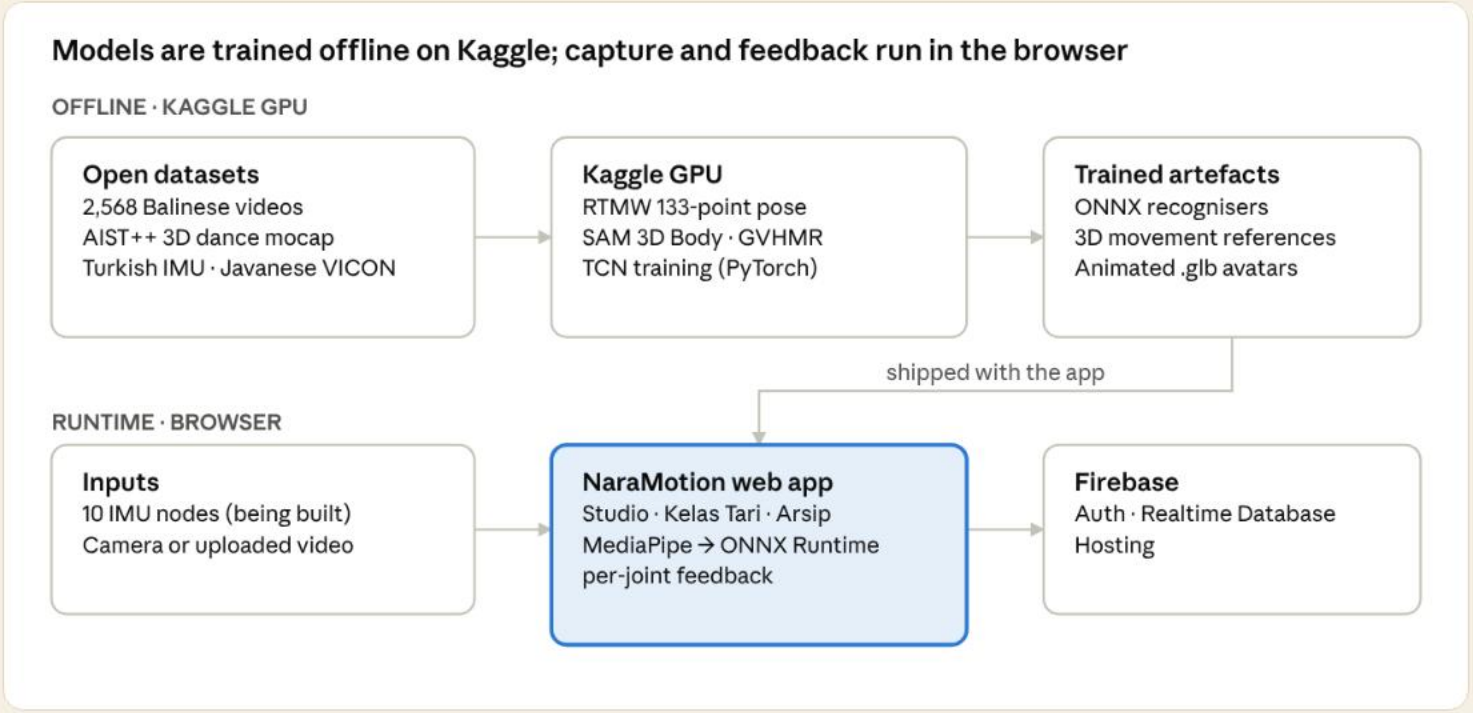


Part	Spec
Nodes	Head, torso, upper/lower arms, thighs, shins
MCU	ESP32-C3 Super Mini
IMU	GY-85: ADXL345, ITG-3205, HMC/QMC5883L
Fusion	Madgwick, 100 Hz
Power	1000 mAh Li-Po, USB-C, OLED
Bandwidth	26 B/sample → 26 kB/s

Suit designed and simulated in the Studio; physical recordings are the next milestone

04 · ARCHITECTURE

Train once in the cloud, run everything in the browser



Offline · Kaggle T4

Pose extraction, TCN training, 3D reconstruction

Runtime · browser

MediaPipe + ONNX Runtime Web; video never leaves the device

Backend · Firebase

Hosting, Auth, Realtime Database for sensor streams

All training used Kaggle's free GPU quota; no paid compute

05 · DATA

Five open datasets, credited by source

2,568
Balinese videos, 24 movements

468,889
frames pose-extracted on GPU

133
keypoints per frame (RTMW)

Dataset	Size	Use
Woman Basic Balinese Dance (CC BY)	1,740 videos · 13 moves · 9 dancers	Recognition, 3D
Male Basic Balinese Dance (CC BY)	828 videos · 11 moves · 4 dancers	Recognition, 3D
AIST++ (CC BY)	411 sequences · 10 genres	Synthetic 10-node IMU
Turkish Folk Dance IMU (CC BY)	30 sessions · 5 real nodes	Sensor recognition
Javanese VICON biomechanics	6 takes · 6 characters	Internal analysis only

06 · AI METHOD

From video to a score in four steps

1 · Pose

RTMW 133 points offline; MediaPipe 17 points live

2 · Normalise

Hip-centred, torso-scaled, 64 frames of x, y, visibility, velocity

3 · Recognise

Dilated TCN, 0.9 M params; male set via transfer + 3-seed ensemble

4 · Score

DTW on 10 segment angles vs 6 exemplars per movement



Browser models: 3.65 MB (female) · 10.93 MB (male ensemble), ONNX opset 17

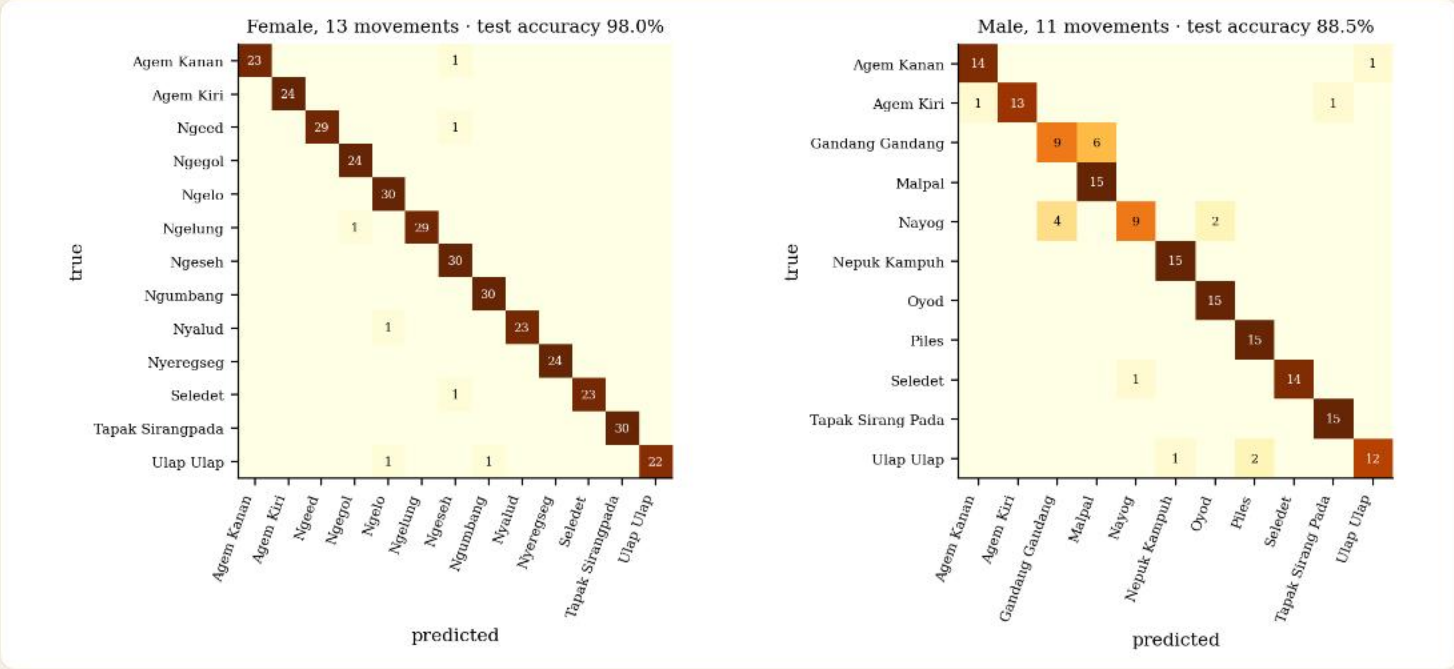
07 · RESULT: RECOGNITION

24 movements recognised on the student's device

98.0%
Female set, 13 moves, browser model

88.5%
Male set, 11 moves: up from 66.7% via transfer + ensemble

95.8%
Male set with all 133 points incl. hands

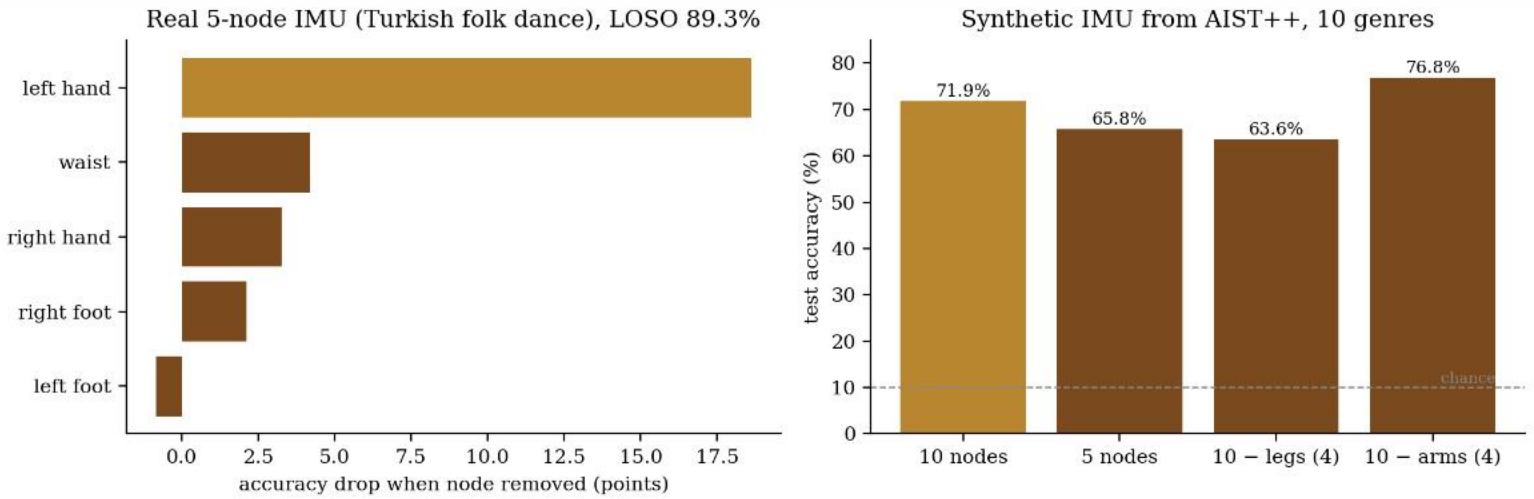


Weakest: Gandang-Gandang and Nayog (60%) → 93.3% and 100% with hand keypoints, so hand tracking is the next upgrade.

Held-out test sets · ST-GCN reference on the female set: 99.4%

08 · RESULT: SENSORS

Limb-segment IMUs carry enough to recognise a dance



89.3%
Real 5-node IMU, unseen dancers (leave-one-subject-out)

−18.6
points when the left-hand node is removed: hands matter most

71.9 vs 65.8%
10 vs 5 synthetic nodes on AIST++ (chance 10%)

Synthetic signals lack real noise and drift; physical-suit validation is pending

09 · RESULT: 3D & CHARACTER

Ordinary video becomes 3D motion, down to the fingers





SAM 3D Body

70 keypoints incl. fingers; 3.9 s/frame on a T4

Motion2MixamoRig

Animated .glb characters in the Arsip

3–4× less jerk

Javanese VICON study: refined *alus* characters move their wrists more slowly and smoothly than *gagah* and animal characters, a measurable style signature.

All 2,568 clips are being reconstructed in 3D on Kaggle (three shards)

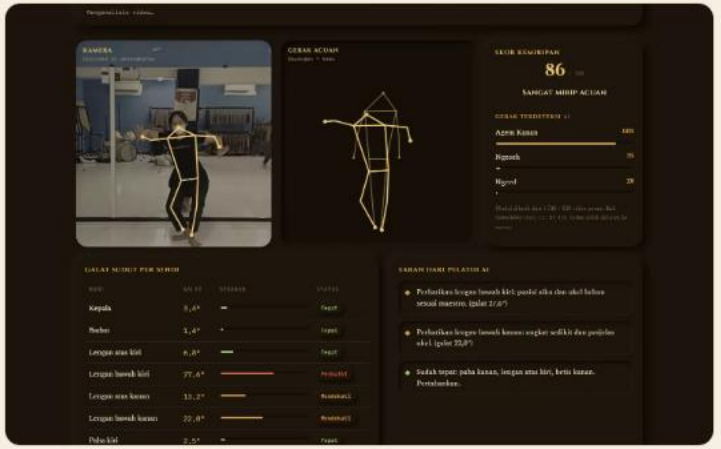
10 · PLATFORM & USE CASES

Three rooms: Studio, Kelas Tari and Arsip



Studio

Replay 3D motion (AIST++, Bali with fingers) and see all 10 sensor streams; record sessions.



Kelas Tari

Camera AI names the movement, scores 0–100 and points to the joints to fix.



Arsip

24 Balinese movements as cards and animated 3D characters, every source credited.

Live now at naramotion.web.app · works on a laptop or phone browser

11 · CONCLUSION & NEXT STEPS

Affordable parts and free compute can keep dance alive as data

Next

1. Build the 10 nodes; record Javanese masters (Gambyong, Serimpi) with synced video
2. 3D scoring from SAM 3D Body references, independent of camera angle
3. Browser hand tracking for finger-level feedback
4. Pilot with sanggar and school classes; measure learning gains

Team

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